



Fig. 2: Experience your dream car

also be run one after another, each of which can be accessed by their run number and run index.

Event-driven design. Not just processes, JaamSim is built to handle event-driven models and a mix of both as well. A method can be processed in parallel or in series, allowing for the much-needed flexibility in system functionality. As a consequence, a new method runs serially and a new process runs in parallel, which means that when a method is halted, the process need not be started all over again.

The entire functioning is managed by Event Manager Object, which maintains separate lists for future events and conditional events, and handles these separately, too. The latest added events are handled first, but conditioned events are processed in the order in which these were entered.

Here, there, everywhere. The one feature you cannot leave out in a real-time system is the factor of dynamic variability. While these cannot be exactly accounted for in simulation, one can certainly consider randomness and statistical phenomena. Here, Probability Distribution feature comes to the aid of the designer.

The designer specifies the range

and time span for such inputs, and these are randomly introduced into the system, paving the way for a nearly-accurate real-time model. JaamSim provides a huge list of distributions to select from, like uniform, triangular, exponential and log-logistic distributions. It also provides user-defined probability distributions, both continuous and discrete. All of these

distributions can also be played around with, like the case of changing the random seed value.

De-cluttering the design maze

JaamSim provides Model Builder and Object Selector to help manoeuvre through the entire system design. While these let you view and select components of the system, each parameter can be customised via Input Editor. Control Panel provides required run control features for simulation. Outputs are mostly pre-programmed depending on the design, but the user can also create outputs for the model using attributes, and even custom outputs for individual blocks. Model outputs can be seen on Output Viewer, while Graphical Window displays a graphical representation of the same.

Look before you leap. Input data has specified formats, and the system can also involve algebraic expressions or Boolean logic. A given input is validated to make sure it satisfies the needed constraints

before being applied to the system. Areas that require a red-green-blue (RGB) colour scheme can use standard colour keywords instead of explicit RGB values. This would be as simple as using pink instead of 255 192 203.

Handling numbers. Numerical details are given in International System of Units (SI units), unless otherwise specified, and a mouse hovering over the output name displays a tool-tip pop-up with a brief description of the selection.

The system being simulated can also have its parameters calculated using various options provided; a controller that signals component updates in the specified sequence, integrators or differentiators, wave generators, delay generators, polynomial calculators, etc.

User interface, the deciding factor

Aesthetics takes an all-new meaning today, and this software keeps up with the trend by letting you change the graphical appearance of the designed model. This feature also comes in handy when moving to designs of higher complexity, as then the placement or appearance might even affect functionality. For example, if a magnet is placed next to a transformer, there is a chance of signal interference.

As you use the software, you can choose between a 2D or 3D view, the former giving a bird's-eye view directly above xy simulation plane and the latter adding a sense of depth in all directions. These views only define the camera position, always giving you full-3D perspective graphics and letting you zoom, pan and orbit over the workspace. Well,

The kind of projects you can simulate with JaamSim

- Construct your dream car from scratch and experience it!
- Create your own server setup.
- How about a harmonic oscillator that models your physical system?
- Or, maybe, a model of the controls in your house!